READING PASSAGE 1

You should spend about 20 minutes on Questions **1–13***, which are based on the Reading Passage 1 below.*

Saffron

Saffron, a spice with a very distinctive bitter taste, notes of hay and metal, and a striking crimson hue, is one of the most highly sought-after commodities in the world, and has been for millennia. *Crocus sativus*, the only crocus flower from which saffron comes, is a member of the iris family that grows perennially from corms, underground plant stems. It is native to southwestern Asia, in and around Greece, and was recorded in a wall painting in Akrotiri in Santorini from about the middle of the second millennium BC. It is also mentioned in an Assyrian 7th century BC botanical treatise. *Crocus sativus* is a delicate flower which grows in woodland and scrub, as well as meadows. Flowering in autumn, it is grown in various regions of the world, primarily along a belt stretching eastwards from the Mediterranean across Iran to India and beyond. It is a sterile plant that does not grow in the wild and needs to be propagated vegetatively, and possibly derives from Crocus cartwightianus, from southwest Asia.

In the world of spices, saffron is by far the most valuable. To help put its value into perspective, a ton of rice in 2017 might cost about $350–360, whereas a kilo of saffron would command between $1,000 and $11,000. The high prices the spice still has today are due in no small part to the fact that it takes about 150,000 crocus flowers to produce about one kilogram of spice. Other factors affecting its worth are the fact that to produce merely one kilogram, a field 2000^2 metres is needed, along with an army of workers to pluck the flowers.

The price of saffron is also influenced by the quality. This, in turn, is dictated by the components of the spice, such as the inclusion of the stamina, the male or pollen-producing parts of a flower, which add bulk but have no effect on flavour. The price is also sensitive to the age of the spice. As the new crop becomes available, bulk buyers of saffron will look for brittleness in the threads and debris at the bottom of the containers, which is an indication of old crop. The colour of the saffron stigmata can also dictate the price, with reddish-brown colour, as opposed to the vibrant crimson, signifying inferior quality.

Crocus sativus that produce saffron grow to 20–30 centimetres in height, and produce up to four flowers of six petals with rounded points. The parts of the flower used to produce the saffron are the female crimson stigmata and their styles, together called threads. The male yellow part of the flower is also harvested and used in the spice, but this produces an inferior product as it does not produce any taste.

Today, *Crocus sativus* is grown commercially in Azerbaijan, Greece, India, Iran, Morocco and Spain, with Iran, the world's largest producer of saffron, accounting for over 90% of the total global yield of approximately 300 tons annually. It is, however, also produced in small quantities, for example in Mund in Switzerland, where about one kilogram a year is produced. At one time in the 16th century, it was even a commercial product in Essex in southeast England, but all that remains is the name, Saffron Walden. A suburb of London, Croydon (from old-English, Grog) a place where crocuses were grown 1,200 years ago, also owes its names to the crocus.

The harvest of this most costly of spices takes place in the autumn. The ground where the crocus flower is grown is kept free of all vegetation. When the flowers appear at the end of October/beginning of November, vast numbers of people are employed, working 24 hours a day in shifts to pluck the flowers by hand, leaving the corm, a bulbous tuber, to flower again the following year. The petals are discarded as waste and the threads or styles are dried traditionally in a saffron kiln. The fresh threads and stigmas need to be dried as a soon as possible to prevent them being ruined by decomposition or mould, and are laid out on mesh screens, which are then baked over hot coals in heated rooms for 10 to 12 hours. The dried spice is subsequently kept in sealed glass containers.

Apart from the culinary uses of saffron, saffron has been used in folk or herbal medicine. In Europe in medieval times, it was used almost as a panacea to treat a vast array of disorders ranging from respiratory problems such as coughs, colds and asthma, as well as smallpox and cancer, blood disorders heart diseases, paralysis, gout and eye disorders. Its yellow colour was also thought of as a cure for jaundice. Among the ancient Egyptians, saffron was used to treat poisoning, and as a digestive and a tonic for dysentery and measles.

It is unlikely that the price or desirability of saffron will decrease, but it is likely that more regions of the world will start to grow the spice, provided the market focuses on its status and value.

Questions 1–5

Do the following statements agree with the information given in Reading Passage 1?

Write:

TRUE *if the statement agrees with the information*

FALSE *if the statement contradicts the information*

NOT GIVEN *if there is no information on this*

1 *Crocus sativus* is grown from seeds.

2 The crocus flower from which saffron comes needs to be cultivated.

3 The price of saffron is governed only by the volume of plants required to produce the spice.

4 The quality of saffron is affected more by age than the inclusion of male stamina.

5 Large quantities of saffron are produced in Switzerland.

Questions 6–10

Complete the flow-chart below.

Choose ***ONE WORD ONLY*** *from the passage for each answer.*

Harvesting saffron

No **6** .... allowed to grow around the crocus flowers

↓

Flowers picked by large numbers of people by **7** .... with the corm being left

↓

Once petals removed, styles are put into a special **8** .... to dry saffron

↓

Done immediately after harvesting to stop fresh threads being **9** ....

↓

Dried spice subsequently kept in airtight **10** .... made of glass

Questions 11–13

Complete each sentence with the correct ending ***A–E****, below.*

11 Iran is responsible for

12 In southeast England, one town has

13 In Ancient Egypt, a cure for poisoning was

A a treatment which contained saffron.

B a district where crocuses are still grown.

C saffron of the best quality.

D the bulk of today's saffron production.

E a reference to saffron in its name.

READING PASSAGE 2

*You should spend about 20 minutes on Questions **14–27**, which are based on Reading Passage 2 below.*

The age of big data

A Humanity has already passed through four main stages of development, namely: the hunter-gatherer, the agricultural, the industrial and the information, each effectively defined by the basic human desire for the acquisition and storage of whatever the focus of the era demands, whether it be food or scientific knowledge. During the information age, vast amounts of data, so-called big data, began to be collected and stored at a phenomenally accelerating rate. Now the world is in a state of flux as it moves to a new era, the fifth stage of human development, dubbed Society 5. This new age is being defined by big data as people seek to harness its power using current advances in technology, including satellite, robotics and cloud computing to solve problems that have until now seemed insurmountable.

B The big data, on which we are now becoming heavily, and perhaps even excessively, dependent, can loosely be defined as vast sets of data about any subject or topic encompassing people's behaviour, as well as factual information, collected and stored at great velocity, and analysed quickly to establish links, patterns and trends. It is now as valuable as any other commodity like petroleum, with practically the same operational complexities involved in working with it. However, it can enhance the decision-making process in a wide variety of fields and organisations.

C Data capture involves a wide range of techniques depending on the field, but the basic principles are the same in each. In the field of weather forecasting, for example, capturing data has moved from radar stations and systems of buoys to the collection of data by satellite, followed by the processing of local, regional, national and international data to provide short-term and long-term forecasts. While weather predictions have been notoriously difficult to make with great accuracy, with the help of big data, the process has improved in recent years as all past weather patterns are now available for forecasters along with current developments.

D Data capture itself is relatively easy compared to the challenges arising from data storage. The global capacity storage has been increasing since the mid-1980s. For example, storage volumes rose daily from 2.6 exabytes (an exabyte being a unit for digital information equivalent to 1 billion gigtabytes) of analog stored data and 0.02 exabytes of data stored digitally, to 19 exabytes and 280 exabytes respectively. This explosion in capacity occurred after the dawn of the digital age in 2002. Today, capacity stands at 35 zettabytes (35 plus 21 zeros, equal to 35,000 exabytes), which is expected to rise to 44 zettabytes of data a year in a few years' time.

E The processing capacity available to manipulate the data is another issue to be contended with. It is far from clear as to whether global data storage centres will be able to deal with the dawn of the internet of things, which according to Gartner, a UK-based technology research centre, will increase to about 26 billion devices in a few years. Moore's law states that data processing speeds will double every two years, but the law is in danger of being broken as processing speeds are declining, with the only hope being developments in quantum physics. This, however, may yet be some time off.

F The impact of big data in a wide range of fields is profound. The internet has provided the opportunity to collect almost unimaginable amounts of data. Researchers now have access to so-called 'found data' created by people using the internet to ascertain, for example, the health behaviour of people to predict the spread of colds and flus. This is like 'quick-and-dirty' data collection as part of a PhD, but it can provide conflicting results.

G In the retail world, data about purchasing patterns can likewise be easily captured. This can be done through loyalty cards for large stores such as supermarkets, which show retailers the items bought, as well as the location and frequency of purchase of any items such as groceries over time. Smartphones connected to roaming wi-fi can also indicate customers' progression through department stores and the exact sections or floors they visited, all of which can give retailers feedback on the efficacy of signage and displays throughout a store. On the internet, purchasing patterns can be stored and even manipulated where, for example, people purchasing an item such as a book or DVD have their attention drawn to other similar purchases made by other site users.

H All this data capture and use may seem slightly sinister because it reduces human knowledge and experience to 'byte size'. It takes away human individuality and privacy. On the bright side, it does and will continue to generate new jobs and to improve the shopping experience.

Questions 14–18

Answer the questions below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

14 Which phase came first as the human race advanced?

15 Which product is big data quoted as being comparable to in worth?

16 What are identical across various fields for collecting data?

17 The beginning of which age led to the rapid increase in storage capacity?

18 How much is the data storage capacity today in exabytes?

Questions 19–22

*Choose the correct letter **A**, **B**, **C** or **D**.*

19 In paragraph A, the writer claims that humanity is about to

A face numerous insoluble problems.

B be overwhelmed by huge amounts of big data.

C move into a new stage of human development.

D leave the industrial age.

20 In the definition of big data, the emphasis when gathering information is on

A links.

B patterns.

C quality.

D speed.

21 The writer claims that big data has

A initiated research projects on weather forecasting.

B enhanced the process of weather prediction.

C affected employee numbers in weather forecasting.

D made shipping forecasts more accurate.

22 According to the writer in paragraph E, it is possible that

A quantum physics will solve the fall in data processing speeds.

B the internet of things will shut down the internet.

C big data storage will continue to double every two years.

D quantum physics is not properly understood.

Questions 23–27

Reading Passage 2 has eight paragraphs, **A–H**.

Which paragraph contains the following information?

Write the correct letter, ***A–H****.*

NB *You may use any letter more than once.*

23 a reference to the creation of employment

24 examples of data capture from people as they shop

25 an evaluation of the effect of big data in different fields

26 a reference to how computer processing speeds may be about to fall

27 how people can be tracked within stores

READING PASSAGE 3

*You should spend about 20 minutes on Questions **28–40**, which are based on Reading Passage 3 below.*

Medicine: time for effective communication

As people live longer and the population increases, the demand for access to medical care is outstripping the funds and time available to National Health Service (NHS) doctors for patient consultations in the UK.

Effective communication in the medical field is essential for the well-being of patients and the efficient running of any hospital and clinic in any health service such as the NHS. In this age of patient-centred communication, time is necessary for the doctor to deal with patients. Such patient-centredness in the doctor-patient consultation puts the patient rather than the doctor, or any other health professional, at the heart of the doctor-patient relationship and apart from the time it demands, it basically involves the establishment of a good rapport and a range of skills to do so.

In a patient-centred approach to history taking, being able to elicit basic information from the patient using questions like open and closed questions is a basic, but invaluable tool in the doctor's repertoire. The latter question type, like *Are you in pain? Is the pain sharp?* requires the patient to answer 'yes' or 'no' to a doctor's enquiry, and may not elicit very helpful information. Open questions, however, put the patient at the heart of the consultation and can provide a much fuller picture of the patient's condition. If the doctor, for example, asks questions such as *What has brought you here?* and the patient answers *I've got a pain, doctor,* the doctor can then ask yes/no questions eliciting information about the site, nature and onset of the pain. If the doctor asks the patient *Can you tell me more about the pain?* the patient is likely to create a holistic narrative describing the pain, for example *I've got this pain here in my right side, which started this morning. It comes and goes. …* This can then, if necessary, be followed by closed questions to elicit and/or clarify specific points. So closed questions with yes/no answers should not be excluded, but used wisely. Question types such as leading or suggestive questions are important to avoid, as the patient may feel obliged to agree with the doctor, and thus give a false picture of their illness.

Apart from questioning, active listening, and hence cue acknowledgment, are also vital in a patient-centred consultation. Active listening means that the doctor listens to the patient and uses all the verbal and non-verbal cues to help reach a diagnosis using their clinical judgment. Verbal cues are what the patient says and non-verbal cues encompass the tone of the patient's voice, eye contact or avoiding eye contact, hesitancy, fidgeting, reluctance to speak, difficulty stopping the patient speaking, and how the patient sits, for example on the edge of the chair or slouched with their head down. These are vital cues as to the patient's feelings, and are all too easy to miss if the doctor is focused on imparting medical judgments in order to race through a tight schedule. For example, restlessness in a patient with a kidney stone might be missed and misdiagnosed to the detriment of the patient and the subsequent cost to the health service.

From the patient's perspective, the benefits of a patient-centred approach in the consultation are clear. It empowers patients, and gives them choices, thus increasing the chances of the patient being diagnosed correctly and quickly, and any treatment such as medication being adhered to. From the medical services point of view, it saves precious time and money, as the patient is likely to be treated properly, thus obviating the need for further diagnostic tests and treatment.

Money is not the answer to every problem, but more investment in the health service is needed, not just to increase the time that patients spend with doctors, but to provide more training for upgrading communication skills in all aspects of the health service. Research on the former shows that in the UK the standard consultation for a General Practitioner (GP) is ten minutes, meaning that GPs can see between 40 and 60 patients a day. This is well above the recommended European average of about 25 patients a day, and below the time for consultation with each patient.

As technology spreads into all aspects of the health service, including the doctor-patient consultation, more research needs to be carried out to ascertain the impact and the limitations of a medical consultation via Skype, or FaceTime. With the emphasis on a patient-centred consultation being on the whole patient, many vital cues may be missed if the doctor is only able to see the patient's face. Skype consultations may have a role to play, but their limitations in a medical consultation context need to be borne in mind, otherwise they could lead to misdiagnosis and greater costs.

Question 28–35

Complete the summary below.

Choose ***NO MORE THAN TWO WORDS*** *from the passage for each answer.*

Elements of effective communication in the medical field

For communication in the medical field to work in the efficiently, being **28** …………………………. when taking a history needs to be adopted. In such an approach simple **29** …………………………. and …………………………. questions are of great value in any medic's **30** …………………………. These, used wisely together, can encourage the patient to provide a **31** …………………………. .
32 …………………………. and …………………………. questions should be avoided as they may end up providing a **33** …………………………. of the patient's condition.
34 …………………………. as well as acknowledging cues are also essential tools as without them the patient's illness might be **35** …………………………. and …………………………. .

Question 36–40

Do the following statements agree with the information given in the Reading Passage?

Write:

YES	*if the statement agrees with the claims of the writer*
NO	*if the statement contradicts the claims of the writer*
NOT GIVEN	*if it is impossible to say what the writer thinks about this*

36 Patient behaviour and attitude can provide the doctor with important information.

37 Putting the patient at the heart of the consultation is cost effective.

38 GPs in the UK spend as much time with patients as their counterparts do in Europe.

39 The use of video technology in the doctor patient consultation could be detrimental to the patient's diagnosis.

40 Research is needed into the cost of providing consultation via video-conferencing.